PROGRAM 5

Working with Vector Database-1 (Working with Collections)

Title: Working with vector database-1(working with collections)

Problem statement: Write a python program to perform following operations on collection

- Create collection
- Add data and display
- Query the collection
- Delete the data

Method:

1. Install chromadb using pip install command and import it.
2. Create chroma client
3. Create collections and perform all operations on collection

Method

Step 1: Check Python Version
Open the VS Code terminal and type:

py --version

Expected output: Python 3.14

Step 2: Install ChromaDB
pip install chromadb

Step 3: Create Python File
Create a file named prg5.py.

Complete Program

```
import chromadb

print("Program started")

# Create ChromaDB client
client = chromadb.Client()

print("Client created")

# Create collection
collection = client.create_collection(name="student_collection")

print("Collection created successfully")


# Add data
print("\n--- Adding Data ---")

collection.add(
    ids=["1", "2", "3", "4"],
    documents=[
        "Python is a programming language",
        "Java is an object oriented language",
        "SQL is used to manage databases",
        "Machine Learning is a branch of Artificial Intelligence"
    ]
)

print("Data added successfully")


# Display data
print("\n--- Displaying Data ---")

data = collection.get()

print(data)


# Query collection
print("\n--- Querying Collection ---")

result = collection.query(
    query_texts=["programming language"],
    n_results=2
)
```

```
print(result)


# Delete data
print("\n--- Deleting Data ---")

collection.delete(ids=["2"])

print("Data with ID 2 deleted successfully")


# Display data after deletion
print("\n--- Data After Deletion ---")

data = collection.get()

print(data)
```

Step 6: Run the Program

python prg5.py

Output

Program started
Client created
Collection created successfully

--- Adding Data ---
Data added successfully

--- Displaying Data ---
[Collection data will be displayed]

--- Querying Collection ---
[Query result will be displayed]

--- Deleting Data ---
Data with ID 2 deleted successfully

--- Data After Deletion ---
[Remaining collection data will be displayed]

Reference

DEPT. OF ISE, CITECH